Holomorphic asymptotics of central shared face coefficients

Abstract

A marked spherical diagram gives an exact coefficient in shared face variables. At each crossing region its finite Casimir polynomial has a central degree part. We prove a nonzero holomorphic Gaussian expansion for the coefficient obtained by selecting that part everywhere, and a strict bound away from the complete spectral point. Both statements hold through all integer levels and both parities, with the original coefficient targets. The local proof starts from the actual cofactor recurrence, including the odd skipped-row term. Positive finite matrices control every other face phase, and the real spectral estimate is sewn on one common radius vector. The resulting estimates supply the source hypotheses of the oscillatory filling theorem. No estimate for the other degree parts of the invariant, or independent identification with a hyperbolic volume, is asserted.

1 The coefficient and the theorem

Let $\mathcal{T}$ be a finite simple oriented triangulation of the sphere. A perfect matching of its dual pairs its triangles across R edges. Its $R+2$ vertices are called faces, following their role in the original diagram. Choose a nonmatching edge uv as the mark. The two face variables L_u, L_v are set to zero; the remaining R variables are ordinary. The order at a matched pair is (x, b, c, y): its triangles are (x, b, y) and (x, c, y), with their induced opposite orientations along the matching edge xy. Repeated occurrences of a face always denote the same variable. In particular $x + y > 0$ on the positive ordinary radii of every pair: otherwise its matching edge would be the unique edge uv, which is the nonmatching mark. Each pair also has a nonmonomial triangle. If one outer face is protected, only one other zero remains, so at least one middle is positive; if both outers are positive, both triangles are nonmonomial.

Fix $N \geq 3$ and write

$$d = N - 1, \quad m = \lfloor N/2 \rfloor, \quad J = \lfloor d/2 \rfloor, \quad \theta = \pi/N, \quad s_j = 2\sin(j\theta), \quad a_j = 4\cos^2((j + \tfrac{1}{2})\theta).$$

The polynomial operator on $1, z, \ldots, z^d$ is

$$\mathsf{C}z^j = 2\cos((2j+1)\theta)z^j - s_j^2(x+y)z^{j-1} - s_j s_{j-1} xyz^{j-2}. \tag{1.1}$$

Absent terms vanish. For $\chi_N(t) = 2T_N(t/2) + 2$, let

$$V_N(\sigma; \mathsf{C}) = \frac{\chi_N(\sigma) - \chi_N(\mathsf{C})}{\sigma - \mathsf{C}}.$$

This is a polynomial divided difference. The finite Fourier theorem [1, Proposition 4.1] identifies C, after evaluation, transpose and a diagonal gauge, with each N-dimensional residue block of the physical pair Casimir. Thus its homogeneous polynomial

$$H_V(x, b, c, y; \sigma) = \sum_{0 \leq i \leq j \leq d} V_N(\sigma; \mathsf{C})_{ij} b^i c^{d-j}$$

retains the original values and nilpotent normals. The central part $H_{V,\mathrm{C}}$ keeps $i \leq J$ and $d - j \leq J$. The other two parts keep $d - j > J$ or $i > J$; we do not estimate them here.

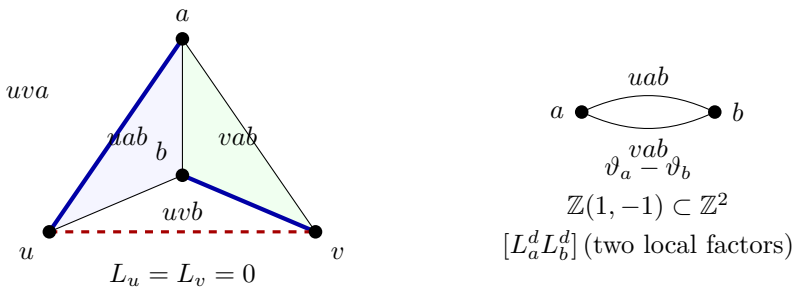


Figure 1: A combinatorial tetrahedral sphere and its ordinary phase graph. The vertices u, v, a, b name shared face variables. The marked edge uv (red, dashed) sets $L_u = L_v = 0$; the two blue matching edges are ua and vb. The outer face of the planar drawing is the triangle uva. The two triangles with two ordinary vertices, uab and vab, compare the same ordinary phases ϑ_a, ϑ_b, giving two copies of the primitive difference row $(1, -1)$. Their coefficients add in the Hessian, while the quotient angular measure remains $d\vartheta_b/(2\pi)$ after $\vartheta_a = 0$. The coefficient target is $L_a^d L_b^d$, with each variable shared by the two local factors. The drawing specifies incidence only; its lengths are not packing radii.

Put $\sigma_e = -2\cos(2\pi\delta_e)$ and define the raw central coefficient

$$C_N(\delta) = \left[\prod_{f \neq u,v} L_f^d\right] \prod_{e=1}^{R} H_{V,\mathrm{C}}(L_{x_e}, L_{b_e}, L_{c_e}, L_{y_e}; \sigma_e)\Big|_{L_u = L_v = 0}. \tag{1.2}$$

This definition includes its sign $(-1)^{dR}$. Its identification as a contribution to the original marked evaluation, with cups, caps and framing still present, is the exact diagram theorem [1, Theorem 9.1 and Corollary 9.2]. The coefficient itself is the object of the present analytic theorem.

Here is the sole positive-radius input. It can be supplied by the complete packing of the marked triangulation, but is stated analytically so that the theorem can be used without a geometric normalization. For nonnegative $(w_1, w_2, w_3) \neq (0, 0, 0)$ define

$$\begin{aligned} \Lambda(t) &= -\int_0^t \log|2\sin v|\, dv, \\ e(a) &= \pi^{-1}\{\Lambda(\pi a) + \Lambda(\pi/2 - \pi a)\}, \\ \mathcal{Q}(w) &= \max_{\substack{a_i \geq 0,\ \sum a_i = 1/2 \\ a_i = 0 \text{ if } w_i = 0}} \sum_i \{e(a_i) + a_i \log w_i\}. \end{aligned} \tag{1.3}$$

Terms on zero coordinates are omitted, and $e(0) = e(1/2) = 0$. Choose positive ordinary radii ρ_f such that

$$\mathcal{F}(L) = \sum_{e=1}^{R} \{\mathcal{Q}(x_e, b_e, y_e) + \mathcal{Q}(x_e, c_e, y_e)\} - \sum_{f \neq u,v} \log L_f \quad \text{is stationary at } L = \rho \tag{1.4}$$

modulo common scale. Set $\Phi = \mathcal{F}(\rho)$. Zeros remain exactly zero. The diagram and radii are fixed before N tends to infinity.

Theorem 1.1 (The actual central source)**.** *Under these hypotheses there is a fixed complex polydisk about zero and, for all sufficiently large N, holomorphic functions A_N and holomorphic $(R-1)$-by-$(R-1)$ matrices H_N on it such that*

$$C_N(\delta) = \frac{(-1)^{dR} N^{2R} e^{N A_N(\delta)}}{(2\pi N)^{(R-1)/2} \sqrt{\det H_N(\delta)}} (1 + O(N^{-1})). \tag{1.5}$$

The relative error is holomorphic and uniform on smaller closed polydisks. The square root continues the positive face Hessian at zero. The actions are separately even and real on the real chart, and $A_N \to A_0$, $H_N \to H_0$ locally uniformly with every fixed derivative. The face Hessian $H_0(0)$ is positive definite. Moreover

$$A_0(0) = \Phi, \qquad A_0''(0) = -2\,\mathrm{diag}(\kappa_e),$$
$$\kappa_e = \frac{\pi\big(\sqrt{b_e(x_e+y_e)+x_ey_e}+\sqrt{c_e(x_e+y_e)+x_ey_e}\big)}{x_e+y_e} > 0. \tag{1.6}$$

All face values in the last expression are at ρ.

For every $0<\eta<1/2$ there are $\gamma_\eta,\tau_\eta>0$ such that, for all sufficiently large integer levels,

$$|C_N(x+\mathrm{i}y)| \le e^{N(\Phi-\gamma_\eta)} \quad (x\in[-1/2,1/2]^R,\ \max_e|x_e|\ge\eta,\ \max_e|y_e|\le\tau_\eta). \tag{1.7}$$

Fixed derivatives have the corresponding bounds on smaller domains. There is no prime-level restriction and no change of normalization between even and odd levels.

We define the finite actions directly from the actual local polynomial. They are not asserted to equal a separately constructed approximate fold interpolation. In particular no value-only interpolation is differentiated to obtain a normal.

2 Finite cofactors and positive nodal matrices

Set $u(\delta) = -4\sin^2(\pi\delta)$, $A = x+y$, $P = xy$ and $\alpha_j = a_j + u$. In this section u is a spectral coordinate, not the protected face. The cleared cofactor is the polynomial

$$R_{-1} = 0, \quad R_0 = 1, \qquad R_j = b^j\prod_{r<j}\alpha_r + s_j^2AR_{j-1} + s_js_{j-1}\alpha_{j-1}PR_{j-2}. \tag{2.1}$$

It is the determinant of the upper two-diagonal matrix with diagonal α_r, lowering entries $-s_r^2A, -s_rs_{r-1}P$, and last row replaced by $(1,b,\dots,b^j)$. Equivalently,

$$R_j = \sum_{i=0}^{j} b^i\prod_{r<i}\alpha_rK_{i,j}, \qquad K_{i,i}=1, \quad K_{i,i-1}=0, \quad K_{i,k} = s_k^2AK_{i,k-1} + s_ks_{k-1}\alpha_{k-1}PK_{i,k-2}. \tag{2.2}$$

These are polynomial identities at a vanishing diagonal too. Write $D_b = R_{m-1}(b,u)$, $E_b = \alpha_{m-1}R_{m-2}(b,u)$ and, at odd levels, $G_b = R_m(b,u)$. A subscript c substitutes c for b. The exact cut of these paths gives [2, Lemma 2.1]

$$P_{\mathrm{C}} := (-1)^dH_{V,\mathrm{C}} = s_m^2AD_bD_c + s_ms_{m-1}P(D_bE_c+E_bD_c), \qquad N=2m, \tag{2.3}$$
$$P_{\mathrm{C}} = G_bG_c + us_m^2PD_bD_c, \qquad N=2m+1. \tag{2.4}$$

The second odd term consists of paths skipping the central row. It is retained throughout, including its first derivative at $u=0$.

We give the finite positive representation needed for phase estimates. Let $S_j = \prod_{r=1}^{j}s_r$, $S_0=1$, and, for $0\le a\le J$, let $H_\pm^{(a)}$ be the Hermitian tridiagonal matrices with

$$(H_\pm)_{n,n} = \frac{1}{\cos(n\theta)\cos((n+1)\theta)}, \qquad 0\le n<a,$$
$$(H_\pm)_{n-1,n} = \sqrt{b_n}\{\sin(n\theta)\pm\mathrm{i}\cos(n\theta)\}, \qquad 1\le n<a, \tag{2.5}$$
$$b_n = \frac{\sin^2(n\theta)}{\cos^2(n\theta)\sin((2n-1)\theta)\sin((2n+1)\theta)}.$$

Empty matrices have determinant one. Put

$$\mathcal{D}_a(x,b,y) = \det[bI + xH_-^{(a)}/2 + yH_+^{(a)}/2]. \tag{2.6}$$

Lemma 2.1 (Exact node adapter)**.** *For $u_a = -a_a$ and $a \le j \le J$,*

$$R_a(b, u_a) = S_{2a}\mathcal{D}_a(x, b, y), \qquad R_j(b, u_a) = K_{a,j}(u_a)R_a(b, u_a). \tag{2.7}$$

Both $H_\pm^{(a)}$ are positive definite. The polynomials $\mathcal{D}_a$ and $K_{a,j}(u_a)$ have nonnegative face coefficients.

Proof. At u_a, the cofactor column on rows $0, \dots, a$ is a null vector of the original triangular matrix. If its coefficients are r_i, its row equations are

$$(a_i - a_a)r_i = s_{i+1}^2 A r_{i+1} + s_{i+2}s_{i+1}P r_{i+2}.$$

The leading coefficient is $\prod_{r<a}(a_r - a_a) = S_{2a}$, by $a_r - a_a = 4\sin((a+r+1)\theta)\sin((a-r)\theta)$. Define $\partial_q b^j = s_j b^{j-1}$. Since $a_i - a_a = s_a s_{a+1} - s_i s_{i+1}$, the normalized monic polynomial is the unique degree-a eigenpolynomial of

$$\mathcal{L} = \partial_q((b^2 + Ab + P)\partial_q), \qquad \lambda_a = s_a s_{a+1}.$$

The eigenvalues through J are distinct: their successive differences are $4\sin(2a\theta)\sin\theta > 0$.

For completeness its monic eigenpolynomials p_a obey

$$p_a = \left(b + \frac{A}{2\cos((a-1)\theta)\cos(a\theta)}\right)p_{a-1} - \frac{b_{a-1}}{4}\{A^2 - 4P\cos^2((a-1)\theta)\}p_{a-2}. \tag{2.8}$$

Here $p_0 = 1, p_1 = b + A/(2\cos\theta)$. One elementary derivation also checks that no lower-degree terms were dropped. Normalize $[j] = s_j/s_1$ and define moments by

$$\mu_0 = 1, \quad \mu_{-1} = 0, \qquad [r+1]\mu_r + A[r]\mu_{r-1} + P[r-1]\mu_{r-2} = 0 \quad (1 \le r \le 2J-1).$$

The sine addition formulas make $\mathcal{L}$ symmetric for $\langle b^i, b^j\rangle = \mu_{i+j}$. Distinct eigenvalues therefore make the p_a orthogonal on the available degrees. Multiplication by b is symmetric as well, so $bp_{a-1} - p_a$ has only the two previous components. Comparison of the two highest coefficients, using the displayed row equations, gives (2.8). Their norm recurrence has its nonzero second coefficient for generic (A, P), starting from norm one. This justifies the induction before polynomial continuation to all A, P; no positive moment assumption is used. All denominators have index less than N. The tridiagonal determinant (2.6) has precisely (2.8), proving the first identity.

The Cholesky pivots of either H are

$$p_{n-1}^H = \frac{\sin(n\theta)}{\cos(n\theta)\sin((2n-1)\theta)} > 0, \qquad 1 \le n \le a.$$

Writing $d_n = (H_\pm)_{nn}$ and $h_n = (H_\pm)_{n-1,n}$, substitution in $d_n - |h_n|^2/p_{n-1}^H = p_n^H$ verifies the factorization, including the last physical row. Thus both matrices are positive. Cauchy–Binet applied to their Cholesky factors proves coefficient positivity of (2.6). For $j > a$, the forcing in (2.1) vanishes and the path must visit a, proving its second identity. On dividing $K_{a,a+r}$ by $\prod_{v=a+1}^{a+r} s_v^2$, the recurrence becomes

$$q_r = (x+y)q_{r-1} - \gamma_r xy q_{r-2}, \qquad \gamma_r = 1 - \frac{s_a s_{a+1}}{s_{a+r}s_{a+r-1}} \in [0, 1].$$

Here $q_0 = 1, q_1 = x + y$. The identity $q_r - xq_{r-1} = y(q_{r-1} - xq_{r-2}) + (1-\gamma_r)xyq_{r-2}$ proves nonnegativity and then $q_r \le (x+y)^r$ coefficientwise. □

At the maximal physical fold the scalar can be read without any interpolation. For $\delta_* = 0$ at odd levels and $\delta_* = 1/(2N)$ at even levels, (2.3)–(2.7) give

$$P_{\mathrm{C}}(\delta_*) = N^2\beta_J^2\mathcal{D}_J(x, b, y)\mathcal{D}_J(x, c, y)(x+y)^{d-2J}, \qquad \beta_J = S_{2J}/S_J^2. \tag{2.9}$$

Indeed $S_{N-1} = N$, $S_J^2 = N$ at odd levels; at even levels $S_{N-2} = N/(2\sin\theta)$, $S_J^2 = N/2$, and $s_m = 2$. Thus both the even forced factor and the odd simple node are literal.

3 Fixed complex cofactor neighborhoods

We next prove the local analytic input for the actual polynomial, independently of its nodal samples. Positive face entries range in a fixed compact set; each zero entry stays zero. Write $n = N/2 + O(1)$, $n \le \lfloor N/2 \rfloor$, for any terminal row used in (2.3)–(2.4). Throughout this section $x + y > 0$, as proved for every actual pair in Section 1. The entropy in (1.3) also allows the different singleton $(0, b, 0)$, but that singleton is not an admissible cofactor chart here.

Proposition 3.1 (Actual local logarithms)**.** *Assume $x, y, b \ge 0$ and $x + y > 0$ at the real reference. On a fixed sufficiently small complex u disk and fixed complex logarithmic neighborhood of the positive faces, every nonzero $R_n(b, u)$ is zero-free for all large N. Its logarithm, continued from positive faces at $u = 0$, satisfies*

$$N^{-1} \operatorname{Log} R_n(b, u) \longrightarrow \psi_b(u) \tag{3.1}$$

locally uniformly, with every fixed parameter derivative. The limit is

$$\begin{aligned} \psi_b(u) &= \phi(t_b(u); u), \\ \phi(t; u) &= t \operatorname{Log} b + \int_0^t \operatorname{Log}(4\cos^2 \pi v + u)\, dv + \int_t^{1/2} \operatorname{Log} \lambda(v, u)\, dv, \\ \lambda(v, u) &= 2 \sin \pi v \left[A \sin \pi v + \sqrt{(x - y)^2 \sin^2 \pi v + (4 + u)P} \right], \\ \sin^2 \pi t_b(u) &= \frac{(1 + u/4)b^2}{(b + x)(b + y)}. \end{aligned} \tag{3.2}$$

For $b = 0$, the expression means $\int_0^{1/2} \operatorname{Log} \lambda$; the sole-variable case is the exact monomer product. All branches continue their positive values at $u = 0$.

3.1 An interior starting row

Suppose $b > 0$. The starting saddle $t_b(0)$ lies strictly between zero and $1/2$. Choose $0 < \ell < t_b(0) < r < 1/2$ with fixed margins, uniformly over the face compact. On a complex neighborhood of $[\ell, 1/2]$, the recurrence for K has coefficients

$$p(t) = 4 \sin^2(\pi t) A, \qquad q(t, h) = 4 \sin(\pi t) \sin(\pi(t - h))[4 \cos^2(\pi(t - h/2)) + u]P, \quad h = N^{-1}.$$

Its dominant characteristic root is a unit, and the ratio of the other root to it has modulus at most $\kappa < 1$. This remains true on fixed sufficiently small complex parameter neighborhoods. A zero upper diagonal can annihilate the smaller root and causes no problem. If $P = 0$ the recurrence is first order.

Here are the needed analytic error estimates. Starting with the characteristic root, iterate

$$r_{k+1}(t) = p(t) + q(t, h)/r_k(t - h)$$

on nested rectangles, advancing their left margins by h. On an $O(h)$ neighborhood of the root, translation changes the root by $O(h)$ and the derivative in the denominator has modulus at most $\kappa' < 1$. After $\lfloor cN \rfloor$ steps, with c smaller than the original margin, the resulting analytic unit r_N satisfies

$$r_N(t) - p(t) - q(t, h)/r_N(t - h) = O(e^{-c_1 N}), \qquad r_N(t) = \lambda(t, u) + O(h). \tag{3.3}$$

One extra shifted rectangle is retained for the residual. The contraction is between nested domains; no translation-invariant analytic norm is assumed. The constants and Cauchy derivative bounds are uniform.

Divide the actual $K_{i,j}$ by $\prod_{v=i+1}^{j} r_N(vh)$. In the zero-residual recurrence its consecutive differences multiply by $-k_N(jh)$, where $k_N(t) = q(t,h)/(r_N(t)r_N(t-h))$ has modulus at most κ'. The actual initial data $K_{i,i} = 1, K_{i,i-1} = 0$ consequently give

$$K_{i,n} = a_N(ih) \prod_{v=i+1}^{n} r_N(vh)(1 + O(e^{-c_2 N})), \tag{3.4}$$

$$a_N(t) = \sum_{l=0}^{\lfloor c_3 N \rfloor} (-1)^l \prod_{v=1}^{l} k_N(t + vh). \tag{3.5}$$

The omitted geometric tail is exponential; a_N tends to $(1 + k_0(t))^{-1}$, a positive unit. The residual in (3.3) adds $O(Ne^{-c_1 N})$ by the same difference recurrence. This proves (3.4) for every actual start $i/N \in [\ell, r]$, with its physical terminal row.

Both logarithmic products in (2.2) admit exponentially accurate analytic primitives on this starting interval. For clarity, the Euler–Maclaurin formula with its periodic Bernoulli remainder [4, §2.10(i)] gives, after $2K$ derivatives, the bound

$$Ch^{2K-1} \frac{(2K)!}{(2\pi d_0)^{2K}} \le Ch^{-1}(C_0 Kh)^{2K}$$

on a retained strip of width d_0. The midpoint formula obeys the same estimate. It follows from the periodic Bernoulli remainder and Cauchy's derivative bound; choose $K = \lfloor c_0/h \rfloor$ with $C_0 c_0 < 1$. Anchor the prefix at zero and the tail at the actual n/N. Thus each starting term is $a_N(i/N)e^{N\phi_N(i/N)}(1 + O(e^{-c_4 N}))$, with analytic finite ϕ_N converging with derivatives to (3.2). All endpoint half terms are retained in ϕ_N.

The omitted starts have a fixed rate loss at positive faces and $u = 0$. Indeed [2, Lemmas 4.1 and 5.2] give the uniform interval upper bound for (2.2), and its action has its unique strict maximum at $t_b(0)$. This loss persists on a fixed small complex parameter neighborhood. One direct quantitative check is to bound each unused diagonal by $a_j + |u|$. The product over distinct used lower rows costs at most $\exp\{N\omega(|u|) + O(\log N)\}$, with $\omega(v) \to 0$: split rows where $a_j \le \sqrt{v}$ (at most $O(Nv^{1/4}) + O(1)$), and use $\log(1 + v/a_j) \le \sqrt{v}$ on the others; the endpoint log sum on the first set is integrable and $O(Nv^{1/4}|\log v|) + O(\log N)$. The odd cofactor never uses its central zero diagonal. A logarithmic face perturbation costs $e^{C\rho N}$ by homogeneous degree. Choose these neighborhoods smaller than the fixed starting-row loss.

We record why the discrete saddle is a contributing one. Surround the kept integer grid by a rectangle with vertical sides on half-grid lines. For $w_N(t) = a_N(t)e^{N\phi_N(t)}$, residues give exactly

$$\sum_i w_N(i/N) = \frac{N}{2\mathrm{i}} \oint \cot(\pi N z) w_N(z)\, dz. \tag{3.6}$$

On the upper and lower sides at fixed height η the cotangent is $\mp\mathrm{i} + O(e^{-2\pi\eta N})$; on the vertical sides its modulus is at most one. The latter retain the endpoint rate loss. The horizontal phase cost at the real reference is $O(\eta^2 N)$, so choose η first with $2\pi\eta > C\eta^2$, then choose the parameter neighborhood. Cauchy's theorem compares (3.6) with N times the actual integral, retaining exponentially smaller vertical currents. The holomorphic implicit function theorem gives its saddle t_N. Deforming to the horizontal line through t_N, with the same endpoint currents, and rescaling by $N^{-1/2}$ yields

$$R_n = \frac{\sqrt{2\pi N}\, a_N(t_N) e^{N\phi_N(t_N)}}{\sqrt{-\phi_N''(t_N)}} (1 + O(N^{-1})). \tag{3.7}$$

The cubic Gaussian term integrates to zero; its square and the quartic term are $O(N^{-1})$. Exponential interpolation errors remain relative errors after the neighborhood is decreased once more. Uniform holomorphy gives the fixed derivative bounds. This proves zero exclusion and (3.1) for $b > 0$.

3.2 A protected middle variable

If $b = 0$ and $xy > 0$, only start zero remains, so it must not be discarded by the preceding interior argument. Set $L_j = R_j/S_j$. Exactly,

$$\begin{pmatrix} L_j \\ L_{j-1} \end{pmatrix} = T_h(jh)\begin{pmatrix} L_{j-1} \\ L_{j-2} \end{pmatrix}, \qquad T_h(t) = \begin{pmatrix} 2\sin(\pi t)A & P[4\cos^2(\pi(t-h/2)) + u] \\ 1 & 0 \end{pmatrix}.$$

The initial vector is $(1,0)^{\mathsf{T}}$. Let $\lambda_\pm(t)$ be the two roots and $g = \lambda_+ - \lambda_-$. The real discriminant at $h = u = 0$ is $4\sin^2(\pi t)A^2 + 16\cos^2(\pi t)P > 0$ on $[0, 1/2]$. Thus g and λ_+ are units on a fixed complex neighborhood. The ratio $r = \lambda_-/\lambda_+$ equals -1 at zero, while $r'(0) = 2\pi A/\lambda_+(0)$ has positive real part. By compactness,

$$|r(t)| \le e^{-ct} \quad (0 \le t \le 1/2), \qquad |1 - r(t)| \ge c.$$

The equality of moduli at the initial point is not an eigenvalue collision.

In the eigenbasis $E(t) = \left(\begin{smallmatrix} \lambda_+ & \lambda_- \\ 1 & 1 \end{smallmatrix}\right)$, let $B_j = E(jh)^{-1}E((j-1)h)$ and $C_j = (B_j)_{++} = (\lambda_+((j-1)h) - \lambda_-(jh))/g(jh)$. Divide out $\prod_{j=1}^n \lambda_+(jh)C_j$. The update is $\mathrm{diag}(1, d_j) + F_j$, with off-diagonal $F_j = O(h)$, $F_j' = O(h)$, $|1 - d_j| \ge c$, and $|d_j| \le e^{Ch - cjh}$. Conjugate by $I + X_j$, where its two off-diagonal entries solve $\mathrm{diag}(1, d_j)X_j - X_j\,\mathrm{diag}(1, d_j) = -F_j$. Then $X_j = O(h)$ and $X_j - X_{j-1} = O(h^2)$, so the new error is $O(h^2)$. All partial products are bounded by e^{Cnh}; telescoping gives an $O(h)$ total error. The negative diagonal product is $O(e^{-c'N})$ for n/N bounded away from zero. The initial eigenbasis coordinates are exactly $(1/g(0), -1/g(0))$. Inverting the final basis proves

$$R_n = \frac{\lambda_+(nh)}{g(0)} \prod_{j=1}^{n} [s_j \lambda_+(jh) C_j](1 + O(N^{-1})). \tag{3.8}$$

Its amplitude is a uniformly bounded unit, since $C_j = 1 + O(h)$. Riemann sums, with the integrable initial $\log\sin$ term, prove (3.1); the analytic relative estimate gives all fixed derivatives. If $b = 0, xy = 0$, the exact answer is $R_n = A^n \prod_{j=1}^n s_j^2$, independent of u. This completes Proposition 3.1.

3.3 The exact central interpolant and its action

The terminal ratios in both proofs tend uniformly to

$$\frac{R_n}{R_{n-1}} \longrightarrow \lambda_* = 2\{A + \sqrt{A^2 + uP}\}.$$

For interior starts the terminal forcing is separated from their saddle and exponentially smaller; for a protected start it follows directly from (3.8). Consequently the exact central factors satisfy

$$\frac{P_{\mathrm{C}}}{D_b D_c} \longrightarrow 4\sqrt{A^2 + uP}\,(N \text{ even}), \qquad \frac{P_{\mathrm{C}}}{G_b G_c} \longrightarrow \frac{2\sqrt{A^2 + uP}}{A + \sqrt{A^2 + uP}}\,(N \text{ odd}). \tag{3.9}$$

These are units near zero, including one-outer-zero cases. Define the exact holomorphic interpolation

$$\mathcal{B}_N(\delta; L) = N^{-2} P_{\mathrm{C}}(u(\delta); L). \tag{3.10}$$

It is zero-free on a fixed chart, is separately even in δ, and is positive on its sufficiently small real positive-face chart. Its physical maximal-fold value is exactly $(2.9)/N^2$. Its logarithm is therefore fixed without a fitted scalar. Equations (3.1) and (3.9) show

$$N^{-1}\,\mathrm{Log}\,\mathcal{B}_N(\delta; L) \longrightarrow Q_b(\delta; L) + Q_c(\delta; L), \tag{3.11}$$

with fixed derivatives. The absolute normalization and derivative are [2, Lemma 5.1]

$$\begin{aligned} Q_b(0) &= \mathcal{Q}(x, b, y), \\ Q_b'(\delta) &= -\operatorname{Log} \frac{E_b(\delta)}{\sqrt{\mathcal{D}(\delta)}}, \quad \mathcal{D} = A^2 - 4P \sin^2 \pi\delta, \\ E_b(\delta) &= A + 2b \sin^2 \pi\delta + 2 \sin \pi\delta \sqrt{bA + P + b^2 \sin^2 \pi\delta}. \end{aligned} \tag{3.12}$$

That lemma proves its additive constant directly from the real cofactor action, using the entropy derivatives and the $b \to \infty$ limit; it makes no volume normalization. Its formula on a real interval equals the limit (3.2), and the identity theorem therefore supplies the same holomorphic branch in (3.11). The constant triangle has $Q_b = \frac{1}{2} \log A$.

4 All face phases and the primitive Gaussian

We prove a global angular bound, not just positive real convexity. For positive semidefinite matrices X_j with $M = \sum X_j > 0$, put $A_j = M^{-1/2} X_j M^{-1/2}$ and $B = \sum e^{\mathrm{i}\vartheta_j} A_j$. The map $z \mapsto (A_j^{1/2} z)_j$ is an isometry, so B is a contraction. Expanding its trace gives

$$|\det B| \le \exp\left[-\frac{1}{2} \sum_{j<k} |e^{\mathrm{i}\vartheta_j} - e^{\mathrm{i}\vartheta_k}|^2 \operatorname{Tr}(A_j A_k) \right]. \tag{4.1}$$

Indeed $\operatorname{Tr}(I - B^*B)$ is the sum displayed without $1/2$, and $\log t \le t - 1$ for the squared singular values. Zero determinants follow by continuity. No commutativity is required.

Lemma 4.1 (Finite triangle phase loss)**.** *For every $a_0 > 0$ and compact positive nonzero-face set, a triangle with at least two positive entries has a phase loss e^{-cN} in $\mathcal{D}_a$ whenever their circular phase diameter is separated from zero, uniformly for $a_0 N \le a \le J$.*

Proof. First let $b > 0$ and use $M = bI + xH_-/2 + yH_+/2$. For each positive outer summand X, its pair trace with bI is $b \operatorname{Tr}(XM^{-2})$. Select every other Cholesky square of H with indices in $[a_0 N/4, a_0 N/2]$. Their normalized vectors w_j have disjoint supports; their pivots and row norms are bounded above and below, and there are at least cN of them. Thus $X \ge c \sum w_j w_j^*$. On these supports $\langle w_j, M w_j \rangle \le C$. Jensen's inequality for the spectral measure of M gives $\langle w_j, M^{-2} w_j \rangle \ge C^{-2}$. Hence each applicable pair trace is at least cN. At three positive entries, one phase difference from b is at least half their diameter. Equation (4.1) proves the claim.

If $b = 0$ and $x, y > 0$, the determinant has the exact expansion

$$\mathcal{D}_a(x, 0, y) = \beta_a^{-1} \sum_{k=0}^{a} \left(\frac{S_a}{S_k S_{a-k}} \right)^2 x^k y^{a-k}, \qquad \beta_a = S_{2a}/S_a^2. \tag{4.2}$$

It follows by substituting the coefficient ratios s_{a-k}^2 / s_{k+1}^2 in the continuant (2.8), with its empty and one-row values. For its positive weights the adjacent ratio is $R(k) = (x/y)[\sin((a-k)\theta)/\sin((k+1)\theta)]^2$ and

$$(\log R)'(t) = \frac{-2\theta \sin((a+1)\theta)}{\sin((a-t)\theta) \sin((t+1)\theta)}.$$

The mode stays cN from both ends. This derivative is at most $-c/N$ everywhere and has absolute value at most C/N near the mode. Multiplication of the ratios bounds the weights above by a Gaussian of width $\sqrt{N}$, and below by a fixed fraction of the modal weight on both shells at distances between $\sqrt{N}$ and $2\sqrt{N}$. The total mass is comparable to $\sqrt{N}$ times the modal weight. Two independent samples from these shells therefore have variance at least cN. The same variance is the log-radius Hessian of (4.2), hence the pair trace in (4.1). This proves the remaining case. A one-variable triangle is a monomial and is explicitly exempt. □

Lemma 4.2 (Terminal cofactor phase loss)**.** *If two positive triangle faces have separated phases, every required R_{J-k}, with k fixed, has an exponential angular loss relative to its positive value at $u = 0$, uniformly for all real $u \in [-4, 0]$ and on a sufficiently thin fixed complex δ strip.*

Proof. We first bound its actual nodes u_a, $a \le J - k$. Choose a small fixed $\delta_0 > 0$. For $a/N \ge 1/2 - \delta_0$, use (2.7) and Lemma 4.1; the remaining K has nonnegative coefficients and its positive value is bounded by its all-monomer product. Both factors are bounded coefficientwise by their $u = 0$ counterparts, as in [2, Lemma 3.1].

For the far nodes $a/N \le 1/2 - \delta_0$, there is a strict loss even without a phase restriction. If $xy > 0$, choose cN disjoint pairs of steps in $[1/2 - \delta_0/2, 1/2 - \delta_0/4]N$. At $u = 0$, replacing either selected monomer pair by its dimer multiplies its weight by at least a fixed positive constant. The independent choices show $K_{a,j}(0) \ge e^{c'N} \prod_{v=a+1}^{j} s_v^2 A$. But $K_{a,j}(u_a)$ is bounded by that monomer product. The product $K_{a,j}(0)R_a(0)$ is an actual subset of paths in $R_j(0)$, so the asserted nodal loss follows. If $xy = 0, b > 0$, write the zero-spectral starts as

$$W_i = b^i \prod_{r<i} a_r \prod_{v=i+1}^{j} s_v^2 A.$$

At a far node, starts $i > a$ vanish, and starts $i \ge \eta N$ cost $\prod_{r<i}(1 - a_a/a_r) \le (1 - c)^{\eta N}$. For small fixed η, the exact ratio $W_{i+1}/W_i = ba_i/(s_{i+1}^2 A)$ is at least $1 + c$ for $i \le 2\eta N$. Every remaining short start is consequently exponentially below $W_{\lfloor 2\eta N \rfloor}$. This covers all nonmonomial zero patterns.

The spectral degree of R_j is at most j. The full cosine-node interpolation on $u_0, \dots, u_J$ has norm $O(\log N)$ [2, Lemma 3.2]. If $j < J$, interpolate instead $R_j(u) \prod_{a=j+1}^{J}(u + a_a)$; its unavailable node values are exactly zero. Removing these k factors costs only a polynomial. Indeed a degree-j polynomial p has $\|p'\| \le C(j + 1)^3 \|p\|$ by its Chebyshev expansion. On a one-sided interval of length $c(j + 1)^{-3}$ next to a maximum, $|p| \ge \|p\|/2$. Delete small neighborhoods of the k roots, of less than half that length. At a remaining point every factor has modulus at least $c_k(j+1)^{-3}$. Division costs $C_k(j+1)^{3k}$. This works for complex coefficients and clustered roots. Thus the nodal loss extends to the whole real interval. Finally, for degree j,

$$|p(-4\sin^2 \pi\delta)| \le e^{2\pi j|\Im\delta|} \|p\|_{[-4,0]}.$$

This is the exterior maximum principle applied to $w^{-j}p(-2 - w - w^{-1})$, $|w| = e^{2\pi|\Im\delta|}$. A small fixed strip preserves half the loss. □

Delete the protected vertices u, v from the ordinary graph. It remains connected: reroute a path around u along its cyclic link, then around v along its link with u removed; the latter is a path or cycle and is connected. Every surviving edge is in a triangle with two ordinary positive entries. No matched edge is uv, so $x + y > 0$ in every box. Each box has at least one nonmonomial triangle: if an outer variable is zero, the other protected variable can kill at most one middle.

The active difference rows $e_f - e_g$ therefore contain a spanning tree of the R ordinary vertices. They span the integer lattice $\{k \in \mathbb{Z}^R : \sum k_f = 0\}$, not a proper sublattice. Fixing one face phase leaves an $(R - 1)$-torus with no additional aliases. If its phase lies outside any neighborhood of zero, some active pair has a separated phase. Every term in (2.3)–(2.4) contains a terminal cofactor from each triangle. Lemma 4.2 therefore gives a loss in that term. All other factors obey the uniform coefficientwise central or cofactor bound [2, Proposition 3.3 and Lemma 3.2]. The finitely many central prefactors compare to $P_{\mathrm{C}}(0)$: use $R_j(0) \ge s_j^2 A R_{j-1}(0)$ and $A^2 \ge 2P$. Consequently, uniformly for every real spectral vector and

a thin strip,

$$\left|\prod_e P_{\mathrm{C},e}(\delta_e;\rho e^{\mathrm{i}\vartheta})\right| \le e^{-cN}\prod_e P_{\mathrm{C},e}(0;\rho) \tag{4.3}$$

on the complementary face torus. Polynomial factors have been absorbed by decreasing c. This proves an estimate for every central summand, not a choice of a favorable one.

5 The finite shared saddle

For the entropy (1.3), the maximizing loads at positive entries are

$$a_i = \pi^{-1}\arctan(w_i/D), \qquad D = \sqrt{w_1w_2 + w_2w_3 + w_3w_1}.$$

They follow from $e'(a) = \log\cot(\pi a)$ and the sum constraint; zeros are interpreted by continuity. Differentiating in logarithmic coordinates gives the positive graph form

$$d^2\mathcal{Q} = \sum_{i<j}\frac{w_iw_j}{2\pi D(w_i+w_j)}(d\log w_i - d\log w_j)^2. \tag{5.1}$$

Terms on zero coordinates are omitted; for a one-variable triangle the Hessian is zero. The ordinary graph connectivity makes the sum positive definite after fixing one ordinary scale. This is an explicit finite-dimensional inference from (1.4), not an assumption about a Gaussian cycle.

Fix one ordinary radius at its reference value and write $L_f = \rho_f e^{z_f}$ for the other $r = R-1$ faces. On the fixed local chart define

$$\Psi_N(z,\delta) = \frac{1}{N}\sum_e \operatorname{Log}\mathcal{B}_{N,e}(\delta_e;L(z)) - \frac{d}{N}\sum_{f\neq u,v}\operatorname{Log}L_f(z). \tag{5.2}$$

All constants, including $O(\log N)/N$ terms and the even fold offset, remain here. Only the explicit N^{2R} is outside this phase. Proposition 3.1 gives convergence with derivatives to

$$\Psi(z,\delta) = \sum_e\{Q_{b_e}(\delta_e;L) + Q_{c_e}(\delta_e;L)\} - \sum_{f\neq u,v}\operatorname{Log}L_f.$$

At $(0,0)$ its z gradient vanishes by (1.4), and its Hessian is positive definite by (5.1). The quantitative holomorphic implicit function theorem now supplies unique nearby $z_N(\delta)$ on one fixed polydisk, converging with derivatives. For example, precondition the gradient by the inverse limiting Hessian: its derivative is uniformly close to the identity on a fixed ball, and its value at zero tends to zero; contraction on a smaller ball proves the assertion uniformly in δ. Set

$$A_N(\delta) = \Psi_N(z_N(\delta),\delta), \qquad H_N(\delta) = \partial_z^2\Psi_N(z_N(\delta),\delta). \tag{5.3}$$

Uniqueness preserves separate evenness and reality.

The coefficient in (1.2) has the exact Cauchy measure $(2\pi)^{-r}d\vartheta$ on this torus. Indeed the numerator has total degree dR, equal to the sum of its ordinary targets; integrating the common phase gives one. Choosing a reference vertex is a primitive integer change of torus variables, so no lattice index multiplies that measure. One may change the radii and phase origins to $\rho e^{z_N(\delta)}$: the integrand is a Laurent polynomial and crosses no pole in the annuli. Near the saddle it is exactly

$$(-1)^{dR}N^{2R}e^{NA_N(\delta)}\exp\left[-\frac{N}{2}\vartheta^{\mathsf{T}}H_N(\delta)\vartheta + O(N|\vartheta|^3)\right].$$

The real part of H_N is uniformly positive definite on a smaller spectral polydisk. The complementary face torus remains exponentially small by (4.3): its real radius changes only

within the fixed positive compact, and its phase origin changes by a small amount. Decrease the spectral chart so this fixed loss exceeds the change in $\Re A_N$. Rescale $\vartheta = N^{-1/2}v$. A fixed small neighborhood is dominated by $e^{-c|v|^2}$; outside it the angular bound applies. Taylor expansion through fourth order, with uniform analytic derivative bounds, gives the Gaussian determinant. The cubic term is odd; its square and the quartic term have relative size $O(N^{-1})$. The remaining Taylor terms and Gaussian tails obey the same bound after a standard $|v| \le N^{1/10}$ split. This proves (1.5), including its coefficient one and its holomorphic relative error.

At zero, mixed z, δ derivatives vanish by separate evenness. Eliminating the face variables therefore does not change the spectral second derivative. Differentiating (3.12) at zero gives

$$Q_b''(0) = -2\pi\sqrt{bA + P}/A.$$

This proves (1.6). Each box has a nonmonomial triangle, so its constant is strictly positive.

6 The entire spectral interval and a fixed strip

The local spectral decrease uses the actual signed transfer, not coefficientwise domination by a positive endpoint. From [2, Lemmas 4.1 and 5.2], for every terminal row $N/2 + O(1)$,

$$\limsup \frac{1}{N} \log |R_j(b, u(\delta))| \le \max\{Q_b(\delta), \tfrac{1}{2}\log b\} \quad (0 \le \delta \le 1/2). \tag{6.1}$$

For $b = 0$ the second branch is omitted. The bound is uniform on the whole interval. By (3.12), Q_b decreases strictly when $b > 0$ or $xy > 0$. Also $Q_b(0) > \frac{1}{2}\log b$ in that case: the endpoint load is not the maximizer of the strictly concave entropy when another entry is positive. The possible endpoint logarithmic singularity at $x = y, \delta = 1/2$ is integrable. Thus a separated $|\delta| \ge \eta$ gives a fixed loss in every nonmonomial triangle.

We specify the small complex face extension used in sewing this bound. In the proof of [2, Lemma 4.1], first remove a small initial interval, then choose a fixed block length so the Cayley–Hamilton bound for each two-dimensional transfer power is within any prescribed exponential error of its spectral radius. Only after choosing that length, perturb the face parameters. Matrix powers over this fixed length vary continuously, so the same block bound persists. The initial interval has the same integrable logarithmic allowance. The scalar unused products are unchanged by face phases. Hence (6.1), with any prescribed error, holds on a fixed sufficiently small complex face neighborhood. This argument neither inverts an eigenbasis at a repeated root nor assumes phase positivity.

Fix the original radii ρ. On the small common-face-phase neighborhood, one separated spectral coordinate supplies the above loss in its nonmonomial triangle; the other factors have their complete upper rates with errors whose finite sum is smaller than half that loss. Equations (2.3)–(2.4) retain both triangle cofactors in each term, and their remaining prefactors are bounded. On the complementary face torus use (4.3), uniform for every real spectral vector. The Cauchy target factor is the same $\prod \rho_f^{-d}$ in both regions. Its complete rate is precisely Φ from (1.4). We have therefore proved a real-cube gap for the shared coefficient, not a sum of separately optimized factors. The same argument without a loss, or the coefficientwise central comparison, gives

$$\sup_{\delta \in [-1/2,1/2]^R} |C_N(\delta)| \le e^{N(\Phi + o(1))}. \tag{6.2}$$

For clarity a final elementary argument supplies a fixed complex strip directly for this shared polynomial. A Laurent polynomial F of degree at most d in each angular coordinate has, in either half-plane of each variable, the subharmonic estimate

$$\log |F(x + \mathrm{i}y)| \le (\mathcal{P}_y * \log |F|)(x) + 2\pi d \sum_e |y_e|. \tag{6.3}$$

Here $\mathcal{P}_y$ is the product of the circle Poisson kernels at radii $e^{-2\pi|y_e|}$; a zero height denotes a point mass. To prove it, multiply by the appropriate z_e^d in each coordinate, apply the disk subharmonic inequality successively, and subtract the known monomial logarithms. Zeros follow by regularization, or by using a continuous upper majorant on the boundary.

Apply the real gap with separation $\eta/2$ and (6.2). If $|x_j| \geq \eta$, the one-coordinate Poisson mass in $|t_j| \geq \eta/2$ tends uniformly to one as $|y_j| \to 0$. Choose the strip so it is at least $3/4$, and so $2\pi R|y|$ and the $o(1)$ term together cost less than a quarter of the real loss. Inequality (6.3) proves (1.7), possibly with a smaller constant. Cauchy's formula on smaller disks and strips gives the fixed derivative assertions. This completes Theorem 1.1.

7 The normalized input to the filling theorem

The hypotheses of [3, Theorem 1.1] now hold for the actual sequence (1.2), with

$$\varepsilon_N = (-1)^{dR}, \qquad \alpha = 2R - (R-1)/2 = (3R+1)/2, \qquad a_N(\delta) = \frac{(2\pi)^{-(R-1)/2}}{\sqrt{\det H_N(\delta)}}. \tag{7.1}$$

The limiting a_N is a nonzero holomorphic function. The finite functions A_N must remain inside the exponential; their convergence does not justify replacing e^{NA_N} by e^{NA_0} in a relative formula.

For fixed independently signed sufficiently large powers n_e, that theorem applies to the original central Fourier period with scalar $\Gamma_N = \exp(\pi \mathrm{i} d^2 \sum n_e/(4N))$. The product of its parity factor and (7.1) is $(-1)^{(N+1)R+dR} = 1$ at every integer level. If z_N is the unit-charge spectral saddle, S_N its finite action, and Q_N its negative spectral Hessian, the resulting leading expression is

$$\Gamma_N(-2\mathrm{i})^R \sqrt{2\pi}\, N^{R+1/2} \frac{\prod_e \sin(2\pi z_{N,e})}{\sqrt{\det H_N(z_N)}\sqrt{\det Q_N}} e^{NS_N}.$$

The relative error and eventual nonvanishing are precisely the conclusions of the oscillatory theorem. Its proof supplies the whole charge sum and endpoint currents; none is inferred from the present face Gaussian alone. Exterior framing or component units, and the modified-dimension factor when that convention is used, remain those of [1].

A geometric companion may identify $2\pi\Phi$ with an augmented volume and $2\pi\Re S_0$ with a specified filled volume. Those statements require its marked peripheral and metric proofs. A full invariant limit also requires control of the complementary degree sectors. Neither conclusion is part of Theorem 1.1.

References

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